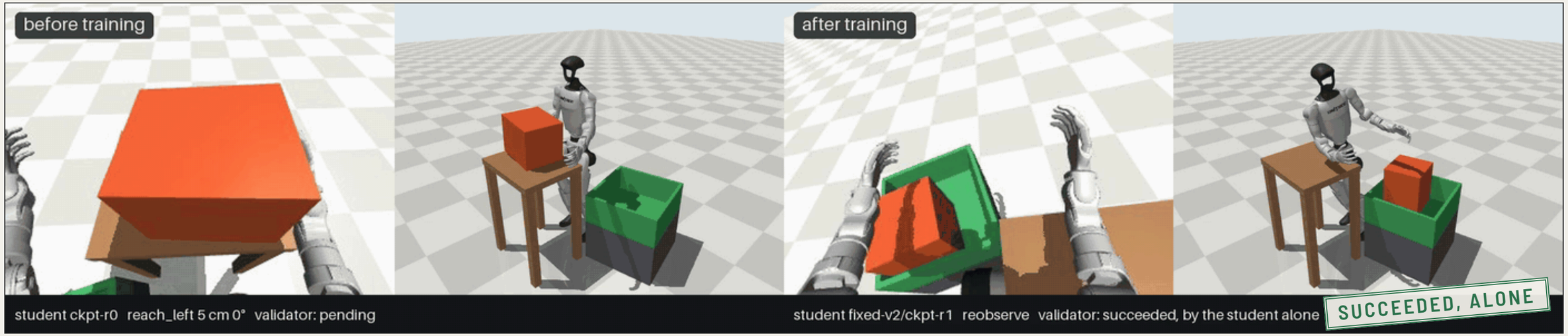
Scaffld

Stop driving robots to teach them. An AI teacher drives them instead.

Exhibit A



Training robots require substantial time and human labour

1

A person drives it

by remote control, recording examples for the robot to copy

→

2

It fails somewhere new

the bin in a new place, the box at a new angle

→

3

They record more

more hours of driving for every new failure

and again, for every task

For DROID, a large robotics research dataset

50 people × 12 months



to record 76k demonstrations by remote control: what robot skills cost at research scale (one cell, one month)

Why now

Humanoids are entering real warehouses.

100,000+ crates

moved by Agility's Digit at GXO's Flowery Branch, GA site; two more US GXO sites from early 2027

Every new job they are given hits this bottleneck.

Scaffld

For a warehouse team, a new task becomes a sentence, not months of teleoperation

Sources: GXO and Agility, 26 Oct 2026, via DC Velocity; DROID, Khazatsky et al., arXiv:2403.12945 v2

Argon writes the task, teaches it, and a validator decides what the robot learns

Steps 3 to 6 repeat: diagnose, teach, fine-tune, exam again

1 — 0 s

Sentence in

“Stay where you are and put the box in the bin beside you, even if someone moves the bin.”

Argon writes the task: 24 s

→

2

Student tries first

The untrained student attempts the sealed scenes alone

Untrained: 0/20

→

3

Diagnose

Argon reads each failed trial: its frames, skill calls and the validator’s reason

10/10 failures diagnosed correctly

→

4 — 15.9 min

Teach

8 Argon teachers demonstrate the fix the diagnosis calls for

The validator kept 56 of 60

→

5 — 21.2 min

Fine-tune

SmolVLM2-500M LoRA, on accepted teacher steps only

605 examples, 2.1 GB peak

→

6 — 2.3 min

Exam again

The student tries again: 20 sealed scenes, no teacher, 0 cloud calls

18/20

39.7 min measured stage time

wall clock; excludes a 70-min GPU queue

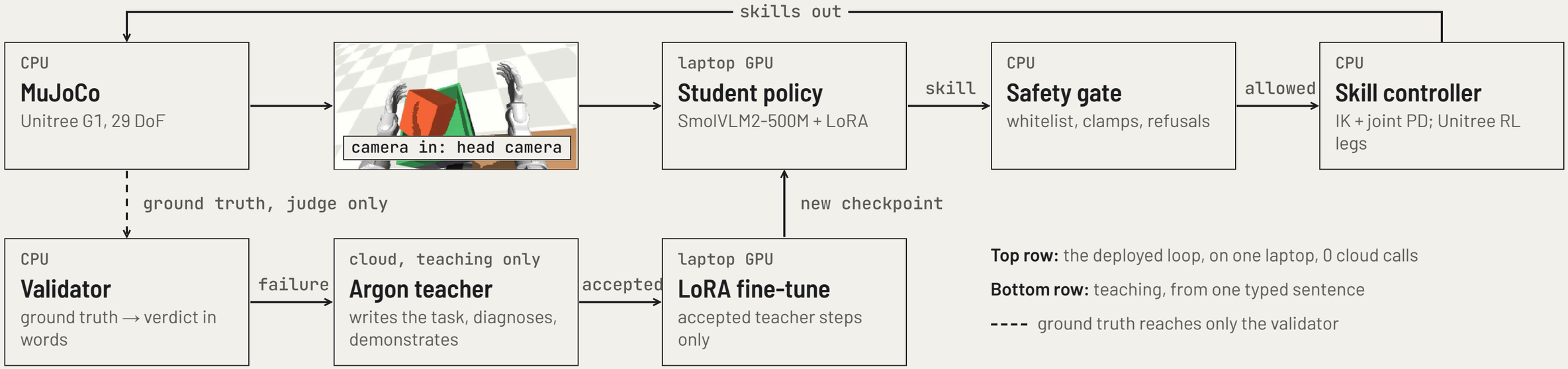


task 24 s · teach 15.9 min · fine-tune 21.2 min · exam 2.3 min

One measured pass. The untrained first try (0/20) ran separately on the same seeds; 10/10 is from repair round 1. Built as an open-source Python pipeline, CLI and live dashboard.

Sources: the showcase run's event logs, PR #34; the untrained run, showcase-compare; repair round 1, PR #10

Camera in, skills out. Ground truth only for the judge



Top row: the deployed loop, on one laptop, 0 cloud calls

Bottom row: teaching, from one typed sentence

---- ground truth reaches only the validator

0.31 s

per student decision on the laptop GPU, 0 cloud calls (n = 326)

2.16 GB of 3 GB

peak student memory of a 3 GB edge budget (v1/v2 exam)

WITHIN BUDGET

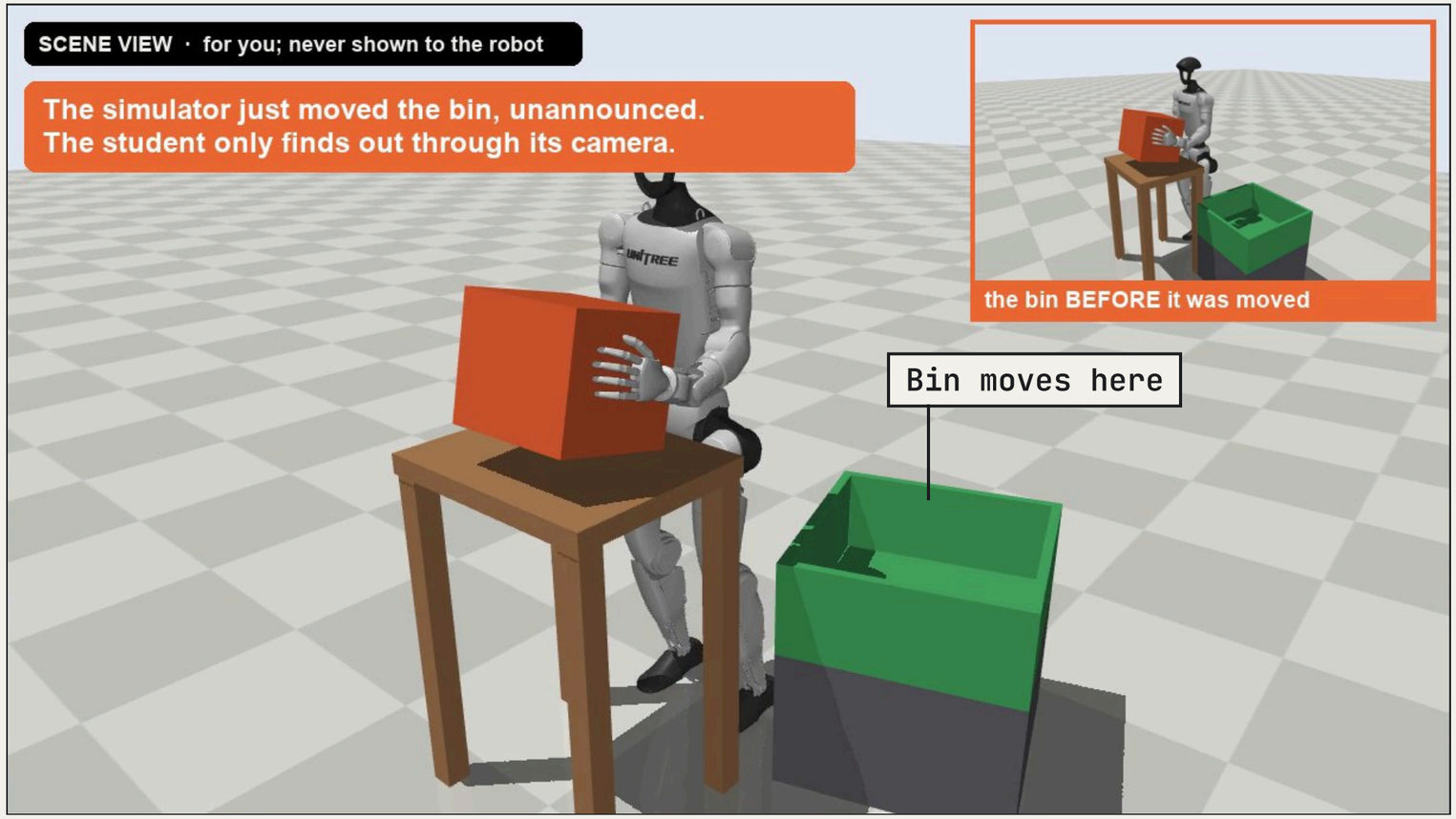
8.1 s

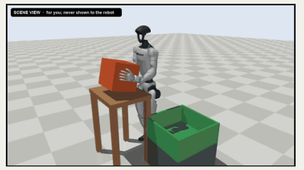
median per Argon call, cloud, teaching only (n = 3,551)

Sources: the repo README and the pitch kit, docs/pitch

Can it still finish when the bin moves?

The trained student acts alone—without help from the teacher.





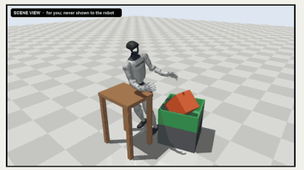
Lift

The robot picks up the box.



Move

We move the bin after the lift.



Finish

The robot puts the box in the bin.

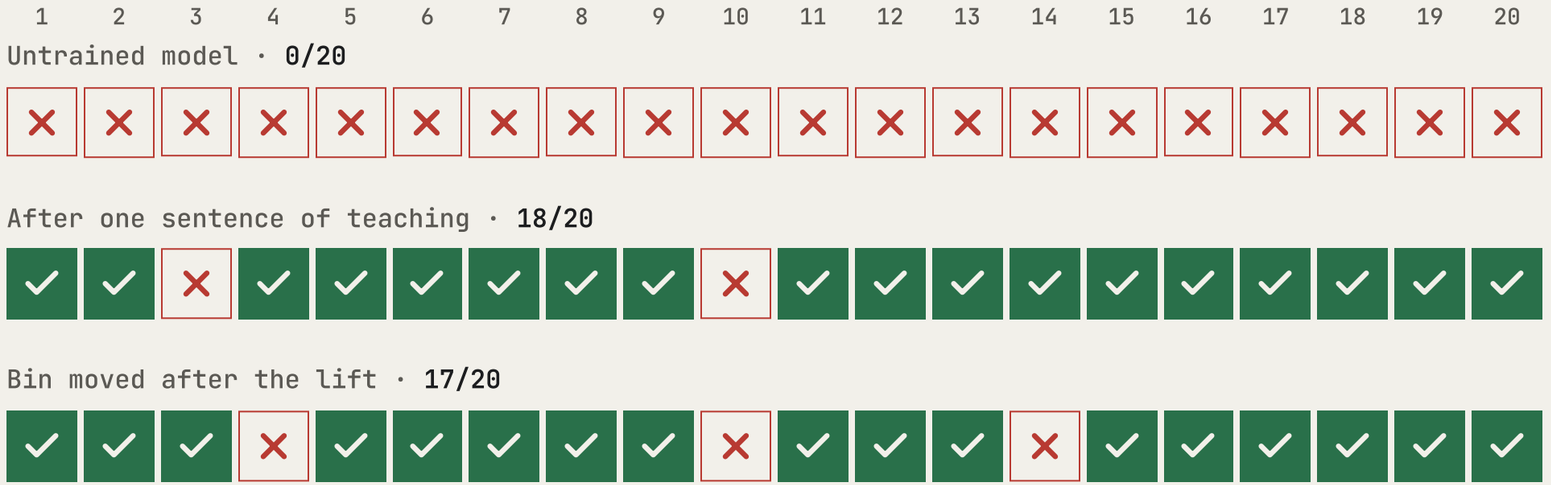
SUCCEEDS

This example succeeds. Across 20 held-out scenes with the bin moved, the student succeeds in 17.

Sources: the 54 s take; 17 of 20 with the bin moved, PR #34

The results before and after training

Laptop GPU, student alone on 20 sealed, unseen scenes



Failed scenes: **3 and 10**; with the bin moved, **4, 10 and 14**. Both: exact McNemar **p < 0.001**, paired by seed; a re-run reproduced every seed.

0/20 → 18/20

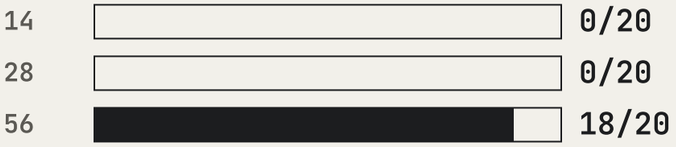
after one sentence of teaching (untrained model 0/20; n = 20)

17/20

with the bin moved after the lift (untrained 0/20)

COPES WITH THE MOVE

Trained on 14, 28 or 56 demos



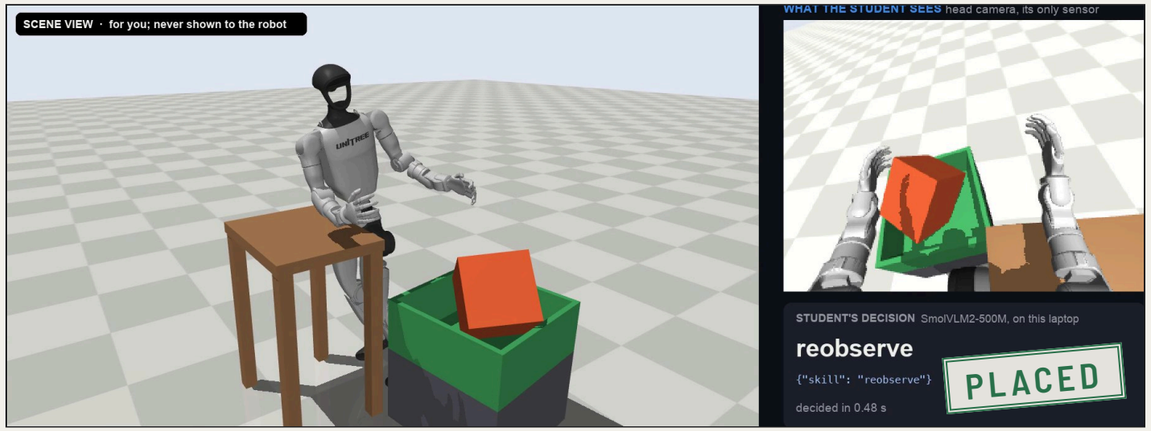
so the demos teach it when to stop

Sources: PR #34 (18/20, 17/20); PR #22 (learning curve); failed seeds from the experiment ledger, docs/pitch

It acts on what it sees

Camera

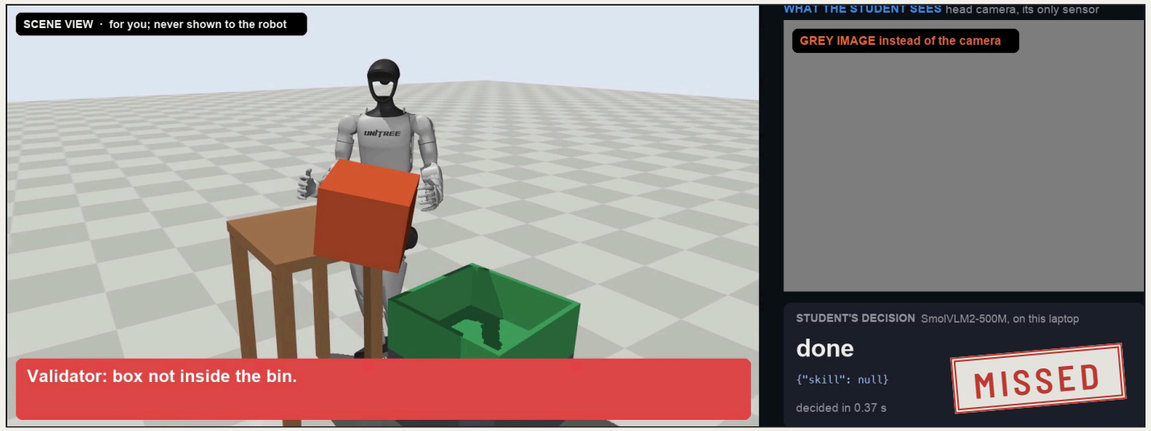
19/20



It twists until the bin is under the box, then places.

Without camera

0/20



Same scene, same student, one twist and it misses.

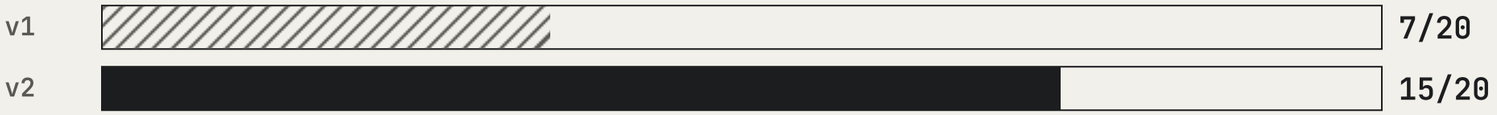
`Control` Same checkpoint, seeds and machine (**Mac**); only the image differs; n = 20 each.

Source: PR #22, camera against a grey image, on a Mac

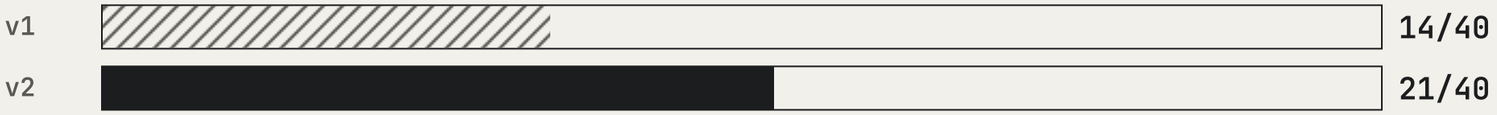
Lessons aimed at failures raise success when things move

Fixed-feet student, v1 → v2, same seeds, student alone

Box moved at the start



Bin placements never seen in training

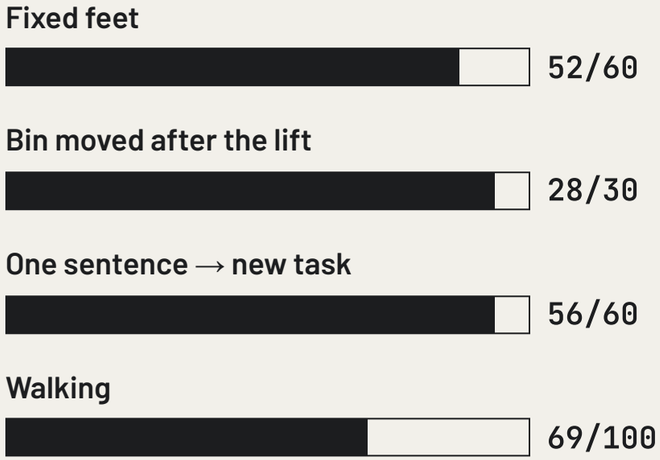


overall p = 0.19; bin further ahead 4/20 → 14/20, p = 0.006

Required caveat

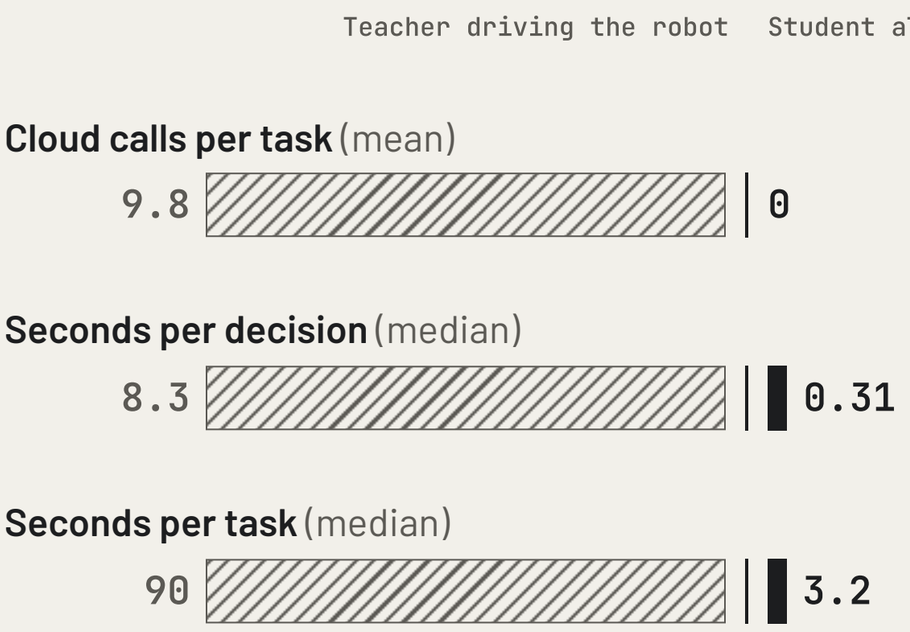
v2 added recovery and box-moved demos (52 → 117), so targeted lessons and more demos aren't separated.

Teacher demos the validator accepted



Sources: PR #22 (v1 against v2); teacher acceptance from the pitch kit, docs/pitch

No humans in the loop. No cloud calls at run time



Teacher on 60 demos, development seeds (**Mac**); student on 40 held-out episodes, 326 decisions (**laptop GPU**).

1 sentence

of human input: 0 teleoperated steps (of 584), 0 labels

20.9 GPU-min

of training on one laptop GPU; 605 examples

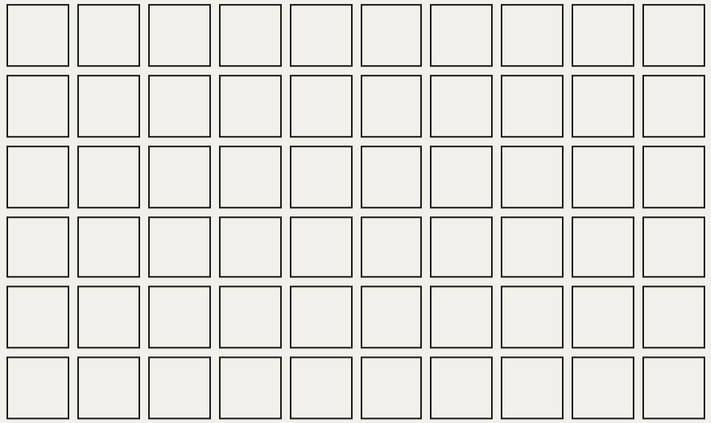
589 Argon calls

for the whole skill, 0 at deployment

Source: the pitch kit, docs/pitch/data/notes.md

Bad lessons are rejected, not used for training

60 teacher demos



Validator

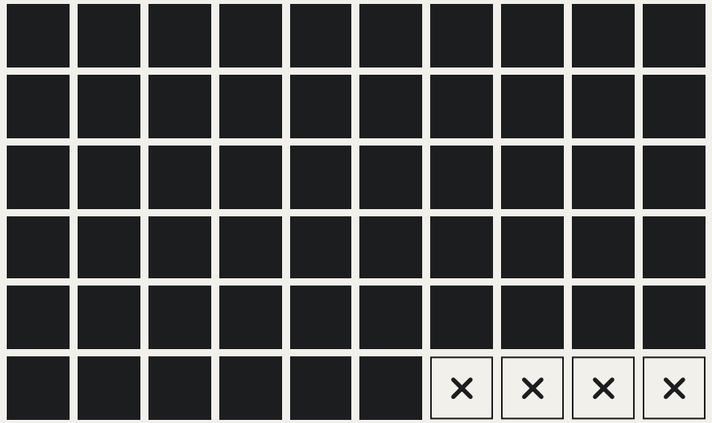
Judges every demo from ground truth, in code, and rejects

falls

joint-limit violations

objects passing through each other

56 accepted for training · 4 rejected



56 accepted for training

4 rejected and never trained on

REJECTED, NOT USED FOR TRAINING

56/56 accepted demos

replay exactly

Unsafe moves

are refused before anything moves

No stale plan

every skill is chosen from the newest camera frame

600+ tests

pin these rules

Sealed test seeds the teacher never sees; an 800-call Argon cap per run; keys never logged · Source: PR #34

What it can't do yet

Limit	Detail	Status
Simulation only	MuJoCo so far, a real robot is next.	SIMULATION
One task family	Box into a bin, learned at the skill level.	SCOPE
No re-aiming yet	1/20 with the bin moved beyond its training range (125°, on a Mac); it keeps its learned routine, while the teacher re-aims.	NOT PASSED
Standing for the loop	The submitted loop runs with fixed feet.	SCOPE
Walking student	2/20 vs 0/20 untrained, not yet significant (n = 20); the teacher walks, 69/100 demos accepted.	NOT YET SIGNIFICANT

Sources: PR #22 (walking 2/20); the bin at 125° from the pitch kit, on a Mac

For a warehouse team, a new task becomes a sentence, not months of teleoperation

Today		With Scaffld
50 people × 12 months of remote-control demos DROID, 76k trajectories	A new task	**1 sentence** and under an hour of compute 39.7 min measured, 0 teleoperated steps

Next

A real Unitree G1

Team

Owen Yan · **Isaac** · **Santi** · **Anh** · **Josh Lim**

github.com/owenbonc/isaac-s-mom

Sources: DROID, arXiv:2403.12945 v2; Scaffld's numbers from saved runs, docs/pitch